AI-GOS Research Pipeline Audit

Complete Section Mapping: Schemas, Tools, MCPs & Output Structures

Date: 2026-03-08 | **Branch**: aigos-v2

Architecture Overview

```
Lead Agent (Claude Opus 4.6 + Vercel AI SDK)
  |
  ├── askUser tool (chip/pill UI questions)
  ├── competitorFastHits (Stage 2, <10s)
  ├── scrapeClientSite (homepage + /pricing)
  ├── confirmJourneyFields (field validation)
  └── generateResearch (dispatches 7 research sections)
        |
        ├── In-process execution (current path)
        |     └── Anthropic SDK agentic loop + betaZodTools
        |
        └── Railway Worker dispatch (legacy path)
              └── Express :3001 → Supabase polling
```

Dispatch pattern: Lead agent calls `generateResearch({ sectionId, context })` → tool runs agentic loop inline → streams chunks to frontend via SSE → persists to Supabase `research_results` JSONB → returns structured result to agent.

Research Dependency Graph

```
Phase 1 (parallel):
  ├── industryResearch      (no deps)
  ├── competitorIntel        (no deps)
  └── icpValidation          (no deps)

Phase 2:
  └── offerAnalysis          (depends: competitorIntel)

Phase 3:
  └── strategicSynthesis     (depends: all 4 above)

Phase 4 (sequential):
  ├── keywordIntel           (depends: strategicSynthesis)
  └── mediaPlan              (depends: strategicSynthesis + keywordIntel)
```

Section 1: Industry Research

Attribute	Value
Key	`industryResearch`
Trigger	`businessModel` confirmed + `primaryIcpDescription` collected
Model	Claude Sonnet 4.6
Max Tokens	8,000
Timeout	180s
Dependencies	None

Tools	`web_search_20250305` (native)
MCPs/betaZodTools	None
Skill	`src/lib/ai/skills/industry-research/SKILL.md`
Runner	`research-worker/src/runners/industry.ts`
Frontend Card	`market-overview-card.tsx`

Output Schema (`src/lib/journey/schemas/industry-research.ts`)

```
{
  categorySnapshot: {
    category?: string
    marketSize?: string
    marketMaturity?: string
    awarenessLevel?: string
    averageSalesCycle?: string
    seasonality?: string
  }
  marketDynamics: {
    demandDrivers: string[]
    buyingTriggers: string[]
    barriersToPurchase: string[]
  }
  painPoints: {
    primary: string[]
    secondary?: string[]
  }
  messagingOpportunities: {
    summaryRecommendations: string[]
  }
  psychologicalDrivers?: { drivers: string[] }
  audienceObjections?: string[]
}
```

Audit Notes

- Uses web search only — no external API enrichment
- Schema is well-structured with clear field names
- `categorySnapshot` fields are all optional — could produce sparse output
- No citation tracking in schema (citations parsed at card level)

Section 2: Competitor Intelligence

Attribute	Value
Key	`competitorIntel`
Trigger	`industryResearch` complete + `topCompetitors` + `websiteUrl` collected
Model	Claude Sonnet 4.6
Max Tokens	8,000+
Timeout	180s
Dependencies	None (but `offerAnalysis` depends on this)

Tools	`web_search_20250305` (native)
MCPs/betaZodTools	`adLibrary`, `spyfu`, `pagespeed`
Skill	`src/lib/ai/skills/competitor-intel/SKILL.md`
Runner	`research-worker/src/runners/competitors.ts`
Frontend Card	`competitor-card.tsx`

betaZodTool Details

Tool	Provider	Input	Output
`adLibrary`	SearchAPI.io (`google_ads_transparency`)	`companyName`, `domain?`	Competitor ad creatives, spend estimates
`spyfu`	SpyFu API	`domain`	Keyword intel, domain organic/paid stats
`pagespeed`	Google PageSpeed Insights (public)	`url` (URL)	Performance score, Core Web Vitals

Output Schema (`src/lib/journey/schemas/competitor-intel.ts`)

```
{
  competitors: Array<{
    name: string
    positioning?: string
    price?: string
    adCount?: number
    strengths: string[]
    weaknesses: string[]
    opportunities: string[]
  }>
  marketStrengths: string[]
  marketWeaknesses: string[]
  whiteSpaceGaps: string[]
  creativeLibrary?: Array<{ summary: string }>
  funnelBreakdown?: Array<{ stage: string, summary: string }>
  keywordOverlap?: number  // 0-100
}
```

Audit Notes

- Richest tool set of all sections — 3 external enrichments
- `adLibrary` depends on `SEARCHAPI_KEY` env var
- `spyfu` depends on `SPYFU_API_KEY` env var
- `pagespeed` uses public API (no key needed)
- `adCount` field could be null if ad library returns no data
- `keywordOverlap` (0-100) is a useful but potentially unreliable metric

Section 3: ICP Validation

Attribute	Value
Key	`icpValidation`
Trigger	`industryResearch` complete + `primaryIcpDescription` collected
Model	Claude Sonnet 4.6
Max Tokens	8,000

Timeout	180s
Dependencies	None
Tools	`web_search_20250305` (native)
MCPs/betaZodTools	None
Skill	`src/lib/ai/skills/icp-validation/SKILL.md`
Runner	`research-worker/src/runners/icp.ts`
Frontend Card	`icp-card.tsx`

Output Schema (`src/lib/journey/schemas/icp-validation.ts`)

```
{
  validatedPersona: string
  demographics: string
  channels: string[]
  triggers: string[]
  objections: string[]
  decisionFactors: Array<{
    factor: string
    relevance?: string
  }>  // min 1 item
  audienceSize: string
  confidenceScore: number  // 0-100
  decisionProcess: string
  coherenceCheck?: string
  painSolutionFit?: string
  riskScores?: Array<{
    factor: string
    relevance?: string
  }>
}
```

Audit Notes

- Web search only — no external API enrichment
- `confidenceScore` (0-100) is model-generated, not empirically validated
- `audienceSize` is a string (e.g., "50,000-100,000") — consider numeric for sorting/filtering
- `riskScores` uses same `{ factor, relevance }` shape as `decisionFactors` — potential naming confusion
- No segment sizing data from external APIs (e.g., LinkedIn audience insights)

Section 4: Offer Analysis

Attribute	Value
Key	`offerAnalysis`
Trigger	`competitorIntel` complete + `productDescription` collected
Model	Claude Sonnet 4.6
Max Tokens	8,000
Timeout	180s
Dependencies	`competitorIntel`

Tools	`web_search_20250305` (native)
MCPs/betaZodTools	`firecrawl`
Skill	`src/lib/ai/skills/offer-analysis/SKILL.md`
Runner	`research-worker/src/runners/offer.ts`
Frontend Card	`offer-analysis-card.tsx`

betaZodTool Details

Tool	Provider	Input	Output
`firecrawl`	Firecrawl API	`url` (URL)	Scraped page content as markdown

Output Schema (`src/lib/journey/schemas/offer-analysis.ts`)

```
{
  offerClarity: {
    clearlyArticulated: boolean
    solvesRealPain: boolean
    benefitsEasyToUnderstand: boolean
    transformationMeasurable: boolean
    valuePropositionObvious: boolean
  }
  offerStrength: {
    painRelevance: number      // 1-10
    urgency: number            // 1-10
    differentiation: number    // 1-10
    tangibility: number        // 1-10
    proof: number              // 1-10
    pricingLogic: number       // 1-10
    overallScore: number       // 1-10
  }
  marketOfferFit: {
    marketWantsNow: boolean
    competitorsOfferSimilar: boolean
    priceMatchesExpectations: boolean
    proofStrongForColdTraffic: boolean
    transformationBelievable: boolean
  }
  redFlags: Array<
    'offer_too_vague' | 'overcrowded_market' | 'price_mismatch' |
    'weak_or_no_proof' | 'no_funnel_built' | 'transformation_unclear'
  >
  recommendation: {
    status: 'proceed' | 'adjust_messaging' | 'adjust_pricing' |
            'icp_refinement_needed' | 'major_offer_rebuild'
    reasoning: string
    actionItems: string[]
  }
  strengths?: string[]
  weaknesses?: string[]
  pricingComparison?: Array<{ competitor: string, price: string }>
}
```

Audit Notes

- `firecrawl` tool scrapes competitor pricing pages — depends on `FIRECRAWL_API_KEY`

- Enum-based `redFlags` is excellent for UI rendering (chips/badges)
- `recommendation.status` enum covers the full decision space well
- `offerStrength` scores are all 1-10 — great for radar charts
- `pricingComparison` is optional — may not populate if Firecrawl fails

Section 5: Strategic Synthesis

Attribute	Value
Key	`strategicSynthesis`
Trigger	All 4 prior sections complete
Model	Claude Sonnet 4.6
Max Tokens	10,000
Timeout	180s
Dependencies	`industryResearch`, `competitorIntel`, `icpValidation`, `offerAnalysis`
Tools	`chartTool` (AntV MCP)
MCPs/betaZodTools	`generateChart`
Skill	`src/lib/ai/skills/strategic-synthesis/SKILL.md`
Runner	`research-worker/src/runners/synthesize.ts`
Frontend Card	`strategy-summary-card.tsx`

betaZodTool Details

Tool	Provider	Input	Output
`generateChart`	AntV MCP Server Chart SDK	`chartType`, `title`, `data[]`, `xField?`, `yField?`, `colorField?`, `valueField?`	Hosted chart image URL

Chart types: `bar`, `pie`, `radar`, `funnel`, `word_cloud`, `dual_axes`, `line`, `sankey`

Output Schema (`src/lib/journey/schemas/strategic-synthesis.ts`)

```
{
  keyInsights: Array<{
    insight: string
    implication: string
    priority: 'high' | 'medium' | 'low'
  }>  // min 1 item
  recommendedPositioning: string
  positioningStrategy: {
    primary: string
    alternatives: string[]
    differentiators: string[]
    avoidPositions: string[]
  }
  recommendedPlatforms: Array<{
    platform: string
    reasoning: string
    priority: 'primary' | 'secondary' | 'testing'
  }>  // min 1 item
  potentialBlockers: string[]
```

```
  nextSteps: string[]
  criticalSuccessFactors?: string[]
  messagingFramework?: {
    coreMessage?: string
    supportingMessages?: string[]
    proofPoints?: string[]
    tonalGuidelines?: string[]
  }
}
```

Audit Notes

- **Bottleneck section** — blocks both `keywordIntel` and `mediaPlan`
- 10,000 max tokens (highest of all sections) — long generation time
- Chart generation adds latency but visual output is high-value
- `messagingFramework` is optional — should be strongly encouraged in skill prompt
- Budget tier rules embedded in system prompt (not configurable)

Section 6: Keyword Intelligence

Attribute	Value
Key	`keywordIntel`
Trigger	`strategicSynthesis` complete
Model	Claude Sonnet 4.6
Max Tokens	4,000
Timeout	180s
Dependencies	`strategicSynthesis`
Tools	None (native)
MCPs/betaZodTools	`spyfu`
Skill	`src/lib/ai/skills/keyword-intel/SKILL.md`
Runner	`research-worker/src/runners/keywords.ts`
Frontend Card	`keyword-intel-card.tsx`

betaZodTool Details

Tool	Provider	Input	Output
`spyfu`	SpyFu API	`domain`	Keyword rankings, CPC, volume, difficulty

Output Schema (`src/lib/journey/schemas/keyword-intel.ts`)

```
{
  keywords: Array<{
    keyword: string
    cpc?: string
    volume?: string | number
    difficulty?: string | number
    opportunity?: number
  }>  // min 1 item
  quickWins: Array<keyword_entry>  // min 1 item
```

```
  highIntentKeywords: Array<keyword_entry>  // min 1 item
  clientStrengths: Array<keyword_entry>  // min 1 item
  contentTopicClusters: Array<{
    theme: string
    keywords: string[]  // min 1 item
    searchVolumeTotal?: string | number
  }>  // min 1 item
  metadata: {
    totalKeywordsAnalyzed: number
  }
}
```

Audit Notes

- Lowest max tokens (4,000) — appropriate for structured keyword lists
- SpyFu is the sole data enrichment — single point of failure for keyword data
- `volume` and `difficulty` accept both `string | number` — normalize before frontend rendering
- `quickWins` vs `highIntentKeywords` vs `clientStrengths` — could overlap; skill prompt should clarify distinction
- No Google Ads Keyword Planner integration (would require OAuth)

Section 7: Media Plan

Attribute	Value
Key	`mediaPlan`
Trigger	`strategicSynthesis` + `keywordIntel` complete + `monthlyAdBudget` collected
Model	Claude Sonnet 4.6
Max Tokens	10,000
Timeout	180s
Dependencies	`strategicSynthesis`, `keywordIntel`
Tools	None (native)
MCPs/betaZodTools	`googleAds`, `metaAds`, `ga4`, `generateChart`
Skill	`src/lib/ai/skills/media-plan/SKILL.md`
Runner	`research-worker/src/runners/media-planner.ts`
Frontend Card	`media-plan-card.tsx`

betaZodTool Details

Tool	Provider	Input	Output
`googleAds`	Google Ads API v18 (OAuth2 + GAQL)	`customerId`, `query?`, `dateRange`	Campaign performance, CPC, CTR, conversions
`metaAds`	Meta Graph API v21.0	`dateRange`	Campaign spend, impressions, conversions
`ga4`	GA4 Data API (JWT service account)	`dateRange`, `focus`	Sessions, audience, channel breakdown
`generateChart`	AntV MCP Server Chart SDK	`chartType`, `title`, `data[]`	Hosted chart image URL

Graceful degradation: All 3 platform tools check credential availability. If unavailable, return `{ available: false, error: '...' }` — agent continues without live data.

Output Schema (`src/lib/journey/schemas/media-plan.ts`)

```
{
  allocations: Array<{
    channel: string
    percentage: number     // 0-100
    spend?: string
    rationale?: string
  }>  // min 1 item
  totalBudget: string
  timeline: string[]
  kpis: Array<{
    channel?: string
    metric?: string
    target?: string
    value?: string
  }>  // min 1 item
  testingPlan: string[]
}
```

Audit Notes

- Richest tool set (4 betaZodTools) but all 3 platform tools require OAuth credentials
- **Most users won't have Google Ads / Meta / GA4 connected** — graceful degradation is critical
- Schema is the leanest of all sections — `timeline` and `testingPlan` are just string arrays
- `allocations.percentage` should sum to 100 — no validation enforced in schema
- `kpis` fields are all optional strings — very loose typing for a critical deliverable
- `totalBudget` is string (e.g., "$5,000/mo") — not machine-parseable

Tool Inventory Summary

betaZodTool Wrappers (8 total)

Tool	Location (Lead)	Location (Worker)	Used By Sections
`firecrawl`	`src/lib/ai/tools/mcp/firecrawl-tool.ts`	`research-worker/src/tools/firecrawl.ts`	offerAnalysis
`adLibrary`	`src/lib/ai/tools/mcp/ad-library-tool.ts`	`research-worker/src/tools/adlibrary.ts`	competitorIntel
`spyfu`	`src/lib/ai/tools/mcp/spyfu-tool.ts`	`research-worker/src/tools/spyfu.ts`	competitorIntel, keywordIntel
`pagespeed`	`src/lib/ai/tools/mcp/pagespeed-tool.ts`	`research-worker/src/tools/pagespeed.ts`	competitorIntel
`generateChart`	`src/lib/ai/tools/mcp/chart-tool.ts`	`research-worker/src/tools/chart.ts`	strategicSynthesis, mediaPlan
`googleAds`	`src/lib/ai/tools/mcp/google-ads-tool.ts`	`research-worker/src/tools/google-ads.ts`	mediaPlan
`metaAds`	*None (worker only)*	`research-worker/src/tools/meta-ads.ts`	mediaPlan
`ga4`	`src/lib/ai/tools/mcp/ga4-tool.ts`	`research-worker/src/tools/ga4.ts`	mediaPlan

Environment Variables Required

Variable	Tool(s)	Required?
`SEARCHAPI_KEY`	adLibrary	Yes (competitor section)
`SPYFU_API_KEY`	spyfu	Yes (competitor + keyword sections)
`FIRECRAWL_API_KEY`	firecrawl	Yes (offer section)
`ANTHROPIC_API_KEY`	All runners	Yes (all sections)
`RAILWAY_WORKER_URL`	Dispatch	Yes (research pipeline)
`RAILWAY_API_KEY`	Worker auth	Yes (research pipeline)
Google Ads OAuth	googleAds	Optional (graceful degradation)
Meta Ads OAuth	metaAds	Optional (graceful degradation)
GA4 Service Account	ga4	Optional (graceful degradation)

Frontend Card Components

Section	Card Component	Key UI Elements
`industryResearch`	`market-overview-card.tsx`	Category snapshot, pain points, demand drivers
`competitorIntel`	`competitor-card.tsx`	Competitor table, strengths/weaknesses, threat levels
`icpValidation`	`icp-card.tsx`	Persona summary, confidence score, triggers, decision factors
`offerAnalysis`	`offer-analysis-card.tsx`	Clarity booleans, strength radar (1-10), red flag badges
`strategicSynthesis`	`strategy-summary-card.tsx`	Insights priority list, platform recs, positioning
`keywordIntel`	`keyword-intel-card.tsx`	Keyword table, quick wins, topic clusters
`mediaPlan`	`media-plan-card.tsx`	Budget allocation, platform breakdown, KPI targets

Common Card Props (`types.ts`)

```
ResearchCardCommonProps<TData> = {
  status: 'streaming' | 'complete' | 'error'
  streamingText?: string
  data?: ResearchCardData<TData>
  citations?: Array<{ number: number, url: string, title?: string }>
  error?: string
  reviewStatus?: 'pending' | 'approved' | 'needs-revision'
  onApprove?: () => void
  onRequestRevision?: (note: string) => void
}
```

Readiness Audit

PASS

Check	Status	Notes
All 7 schemas defined	PASS	Zod schemas in `src/lib/journey/schemas/`
All 7 runners exist	PASS	`research-worker/src/runners/`

All 7 frontend cards	PASS	`src/components/journey/research-cards/`
Dependency graph enforced	PASS	System prompt trigger rules + `previousSections`
Dispatch with retry	PASS	3x retry with exponential backoff
Job status tracking	PASS	`job_status` JSONB column in Supabase
Error states handled	PASS	`data-research-status` with `status: 'error'`
Graceful degradation	PASS	Google/Meta/GA4 tools return `available: false`
Section-to-key mapping	PASS	`research-worker/src/section-map.ts`
Citation parsing	PASS	`research-output.ts` handles `[n]: url` format

WARNINGS

Check	Status	Notes
metaAds missing from lead agent	WARN	Only exists in worker — lead agent `competitorFastHits` can't use it
Schema type inconsistencies	WARN	`volume`/`difficulty` accept `string | number` in keyword schema
mediaPlan kpis loosely typed	WARN	All fields optional strings — could produce empty KPI objects
`allocations.percentage` no sum validation	WARN	Could produce allocations that don't sum to 100%
`totalBudget` not machine-parseable	WARN	String format ("$5,000/mo") blocks programmatic use
`categorySnapshot` all-optional	WARN	Could produce empty snapshot if model doesn't populate
Dual implementations	WARN	Both lead agent and worker have tool wrappers — maintenance burden

BLOCKERS (must fix before production)

Check	Status	Notes
`RAILWAY_WORKER_URL` required	BLOCKER	Without it, all research dispatches silently fail
`ANTHROPIC_API_KEY` required	BLOCKER	All runners need this
Supabase connectivity	BLOCKER	Worker writes + lead agent reads from same table
Worker health endpoint	CHECK	Verify `/health` returns 200 before dispatch

---

Recommendations

1. **Tighten mediaPlan schema**: Make `kpis` fields required, add `percentage` sum constraint
2. **Normalize keyword types**: Convert `string | number` to `number` with parser
3. **Add metaAds to lead agent MCP wrappers**: Parity with worker toolset
4. **Add schema version field**: Allow backward-compatible schema evolution
5. **Monitor tool timeout rates**: 180s timeout could be tight for multi-tool runners
6. **Add Supabase health check**: Verify connectivity before starting research pipeline

---

Part 2: V1 → V2 Schema Comparison & Elevation Plan

V1 Section Names → V2 Section Names

V1 Key	V2 Key	Status

`industryMarketOverview`	`industryResearch`	Renamed + simplified
`competitorAnalysis`	`competitorIntel`	Renamed + heavily simplified
`icpAnalysisValidation`	`icpValidation`	Renamed + heavily simplified
`offerAnalysisViability`	`offerAnalysis`	Renamed + slightly simplified
`crossAnalysisSynthesis`	`strategicSynthesis`	Renamed + simplified
`keywordIntelligence`	`keywordIntel`	Renamed + heavily simplified
(separate 10-section pipeline)	`mediaPlan`	New — collapsed from 10 sections to 1

Section-by-Section Comparison

1. Industry & Market

V1 (`industryMarketOverview`) — 6 top-level objects, deeply structured:

```
{
  categorySnapshot: {
    category: string                      // REQUIRED
    marketMaturity: 'early' | 'growing' | 'saturated'  // ENUM
    awarenessLevel: 'low' | 'medium' | 'high'          // ENUM
    buyingBehavior: 'impulsive' | 'committee_driven' | 'roi_based' | 'mixed'  // ENUM
    averageSalesCycle: string
    seasonality: string
  }
  marketDynamics: {
    demandDrivers: string[]               // 4-6 items
    buyingTriggers: string[]              // 4-6 items
    barriersToPurchase: string[]          // 3-5 items
    macroRisks: {                         // ← V2 MISSING
      regulatoryConcerns: string
      marketDownturnRisks: string
      industryConsolidation: string
    }
  }
  painPoints: {
    primary: string[]                     // 5-7 items
    secondary: string[]                   // 5-8 items (REQUIRED in V1)
  }
  psychologicalDrivers: {                 // ← V2 has this as optional string[]
    drivers: Array<{
      driver: string
      description: string                 // ← V2 MISSING: no descriptions
    }>
  }
  audienceObjections: {                   // ← V2 has this as optional string[]
    objections: Array<{
      objection: string
      howToAddress: string                // ← V2 MISSING: no response strategies
    }>
  }
  messagingOpportunities: {
    summaryRecommendations: string[]      // 3 items
  }
}
```

V2 (`industryResearch`) — Same structure but with key losses:

```
{
  categorySnapshot: {
    category?: string                    // ALL OPTIONAL (was required)
    marketSize?: string                  // NEW field
    marketMaturity?: string              // Downgraded from enum to string
    awarenessLevel?: string              // Downgraded from enum to string
    averageSalesCycle?: string
    seasonality?: string
    // MISSING: buyingBehavior enum
  }
  marketDynamics: {
    demandDrivers: string[]
    buyingTriggers: string[]
    barriersToPurchase: string[]
    // MISSING: macroRisks object
  }
  painPoints: {
    primary: string[]
    secondary?: string[]                 // Downgraded to optional
  }
  messagingOpportunities: {
    summaryRecommendations: string[]
  }
  psychologicalDrivers?: { drivers: string[] }  // Flattened, no descriptions
  audienceObjections?: string[]                  // Flattened, no response strategies
}
```

Gap Analysis:

Lost from V1	Impact	V3 Recommendation
`buyingBehavior` enum	Loses ad targeting signal	**Restore** — critical for campaign structure
`macroRisks` object	Loses regulatory/downturn context	**Restore** — affects budget planning
`psychologicalDrivers.description`	Loses "how it manifests" detail	**Restore** — drives creative copy
`audienceObjections.howToAddress`	Loses response strategy per objection	**Restore** — drives objection-handling ads
Enum types → strings	Loses frontend rendering precision	**Restore enums** — enables chips/badges
Required → optional fields	May produce sparse output	**Make `category`, `marketMaturity` required**
`marketSize` field	V2 has this (NEW)	**Keep** — good addition

---

2. Competitor Intelligence

V1 (`competitorAnalysis`) — Deeply structured per competitor:

```
{
  competitors: Array<{
    name: string
    website?: string
    positioning: string
    offer: string                        // ← V2 MISSING
    price: string
    funnels: string                      // ← V2 MISSING
    adPlatforms: string[]                // ← V2 MISSING
```

```
    strengths: string[]
    weaknesses: string[]
    threatAssessment?: {                // ← V2 MISSING (entire object)
      threatFactors: {
        marketShareRecognition: number   // 1-10
        adSpendIntensity: number         // 1-10
        productOverlap: number           // 1-10
        priceCompetitiveness: number     // 1-10
        growthTrajectory: number         // 1-10
      }
      topAdHooks?: string[]
      likelyResponse?: string
      counterPositioning?: string
    }
    pricingTiers?: Array<{              // ← V2 MISSING
      tier: string
      price: string
      features?: string[]
    }>
    reviewData?: {                      // ← V2 MISSING
      trustpilot?: { trustScore, totalReviews, aiSummary }
      g2?: { rating, reviewCount, productCategory }
    }
  }>
  creativeLibrary: {                    // V1: structured booleans
    creativeFormats: {
      ugc: boolean
      carousels: boolean
      statics: boolean
      testimonial: boolean
      productDemo: boolean
    }
  }
  funnelBreakdown: {                    // V1: structured analysis
    landingPagePatterns: string[]
    headlineStructure: string[]
    ctaHierarchy: string[]
    socialProofPatterns: string[]
    leadCaptureMethods: string[]
    formFriction: 'low' | 'medium' | 'high'
  }
  marketStrengths: string[]
  marketWeaknesses: string[]
  whiteSpaceGaps: Array<{             // V1: structured objects
    gap: string
    type: 'messaging' | 'feature' | 'audience' | 'channel'
    evidence: string
    exploitability: number              // 1-10
    impact: number                      // 1-10
    recommendedAction: string
  }>
}
```

V2 (`competitorIntel`) — Dramatically simplified:

```
{
  competitors: Array<{
    name: string
    positioning?: string                // Downgraded to optional
```

```
    price?: string                    // Downgraded to optional
    adCount?: number                  // NEW (replaces adPlatforms)
    strengths: string[]
    weaknesses: string[]
    opportunities: string[]           // NEW (replaces threatAssessment)
    // MISSING: offer, funnels, adPlatforms, threatAssessment,
    //          pricingTiers, reviewData
  }>
  marketStrengths: string[]
  marketWeaknesses: string[]
  whiteSpaceGaps: string[]            // Downgraded from objects to strings
  creativeLibrary?: Array<{ summary: string }>  // Flattened from structured booleans
  funnelBreakdown?: Array<{ stage: string, summary: string }>  // Flattened
  keywordOverlap?: number             // NEW (0-100)
}
```

Gap Analysis:

Lost from V1	Impact	V3 Recommendation
`threatAssessment` (5-factor scoring)	Loses quantified threat ranking	**Restore** — core competitive analysis
`offer`, `funnels`, `adPlatforms`	Loses go-to-market intelligence	**Restore** — critical for positioning
`pricingTiers` (from Firecrawl)	Loses pricing page intelligence	**Restore** — drives pricing strategy
`reviewData` (Trustpilot, G2)	Loses social proof intelligence	**Restore** — drives proof-point ads
`creativeLibrary.creativeFormats` booleans	Loses structured creative intel	**Restore** — determines ad format strategy
`funnelBreakdown` structured analysis	Loses landing page + CTA patterns	**Restore** — informs funnel design
`whiteSpaceGaps` structured objects	Loses exploitability/impact scoring	**Restore** — quantifies opportunity size
`keywordOverlap` + `adCount`	V2 additions	**Keep** — useful additions

---

3. ICP Validation

V1 (`icpAnalysisValidation`) — The most comprehensive V1 section (12 top-level fields):

```
{
  coherenceCheck: {                   // ← V2: flattened to optional string
    clearlyDefined: boolean
    reachableThroughPaidChannels: boolean
    adequateScale: boolean
    hasPainOfferSolves: boolean
    hasBudgetAndAuthority: boolean
  }
  painSolutionFit: {                  // ← V2: flattened to optional string
    primaryPain: string
    offerComponentSolvingIt: string
    fitAssessment: 'strong' | 'moderate' | 'weak'
    notes: string
  }
  marketReachability: {               // ← V2 MISSING entirely
    metaVolume: boolean
    linkedInVolume: boolean
    googleSearchDemand: boolean
    contradictingSignals: string[]
  }
  economicFeasibility: {              // ← V2 MISSING entirely
```

```
    hasBudget: boolean
    purchasesSimilar: boolean
    tamAlignedWithCac: boolean
    notes: string
  }
  riskScores: Array<{                 // V1: full risk matrix
    risk: string
    category: 'audience_reachability' | 'budget_adequacy' | 'pain_strength' |
              'competitive_intensity' | 'proof_credibility' | 'platform_policy' |
              'seasonality' | 'data_quality'
    probability: number                // 1-5
    impact: number                     // 1-5
    earlyWarningIndicator?: string
    mitigation?: string
    contingency?: string
    budgetImpactEstimate?: string
  }>
  finalVerdict: {
    status: 'validated' | 'workable' | 'invalid'
    reasoning: string
    recommendations: string[]
  }
  customerPsychographics: {            // ← V2 MISSING entirely
    goalsAndDreams: string[]
    fearsAndInsecurities: string[]
    embarrassingSituations: string[]
    perceivedEnemy: string
    failedSolutions: string[]
    dayInTheLife: string               // 1st-person narrative
  }
  triggerEvents: Array<{               // ← V2: flattened to string[]
    event: string
    annualFrequencyEstimate: string
    urgencyLevel: 'immediate' | 'near-term' | 'planning-cycle'
    detectionMethod: string
    recommendedHook: string
  }>
  segmentSizing: Array<{              // ← V2 MISSING entirely
    totalAddressableAccounts: number
    totalAddressableContacts: number
    segmentSharePercent: number
    priorityTier: number
    recommendedBudgetWeight: number
    priorityFactors: {
      painSeverity: number             // 1-10
      budgetAuthority: number          // 1-10
      reachability: number             // 1-10
      triggerFrequency: number         // 1-10
    }
  }>
  samEstimate: {                       // ← V2 MISSING entirely
    totalMatchingCompanies: number
    filteringFunnel: Array<{ stage, count, dropOffReason }>
    estimatedSAMCompanies: number
    estimatedAnnualContractValue: number
    confidence: 'high' | 'medium' | 'low'
    dataSources: string[]
  }
  sensitivityAnalysis: {               // ← V2 MISSING entirely
    bestCase: { assumedCPL, leadToSqlRate, sqlToCustomerRate, conditions }
```

```
    baseCase: { assumedCPL, leadToSqlRate, sqlToCustomerRate, conditions, confidencePercent }
    worstCase: { assumedCPL, leadToSqlRate, sqlToCustomerRate, conditions }
    breakEven: { maxCPLFor3xLTV, maxCAC, minLeadToSqlRate, budgetFloorForTesting }
  }
}
```

V2 (`icpValidation`) — Drastically simplified:

```
{
  validatedPersona: string
  demographics: string
  channels: string[]
  triggers: string[]                    // Flattened from structured objects
  objections: string[]
  decisionFactors: Array<{ factor: string, relevance?: string }>
  audienceSize: string
  confidenceScore: number               // 0-100
  decisionProcess: string
  coherenceCheck?: string               // Flattened from 5-boolean object
  painSolutionFit?: string              // Flattened from structured object
  riskScores?: Array<{ factor: string, relevance?: string }>  // Simplified
}
```

Gap Analysis:

Lost from V1	Impact	V3 Recommendation
`coherenceCheck` 5-boolean structure	Loses paid channel viability signals	**Restore** — go/no-go for ad spend
`painSolutionFit` structured object	Loses fit assessment enum + notes	**Restore** — affects offer messaging
`marketReachability` entire object	Loses Meta/LinkedIn/Google volume checks	**MUST restore** — core platform viability
`economicFeasibility` entire object	Loses budget/TAM/CAC alignment	**MUST restore** — determines if ads viable
`customerPsychographics` entire object	Loses emotional drivers, day-in-life narrative	**MUST restore** — gold for ad creative
`triggerEvents` structured objects	Loses frequency, urgency, detection method, hooks	**Restore** — drives trigger-based campaigns
`segmentSizing` entire object	Loses TAM/contacts/budget weight per segment	**Restore** — determines budget allocation
`samEstimate` entire object	Loses filtering funnel + ACV estimate	**Restore** — ROI projection input
`sensitivityAnalysis` entire object	Loses best/base/worst case CAC projections	**MUST restore** — core financial viability
`riskScores` probability × impact matrix	Loses quantified risk assessment	**Restore** — risk-based budget guardrails

This is the biggest regression. V1 ICP was ~12 structured objects; V2 is ~12 flat strings/arrays.

---

4. Offer Analysis

V1 (`offerAnalysisViability`): Nearly identical to V2.

Delta: V1 schema and V2 schema are **~95% the same**. The V2 additions of `strengths`, `weaknesses`, and `pricingComparison` are improvements.

Changed	V1 → V2	V3 Recommendation
`strengths`	NEW in V2	**Keep**
`weaknesses`	NEW in V2	**Keep**
`pricingComparison`	NEW in V2	**Keep**

Verdict: V2 offer analysis is the closest to V1 parity. Minor elevation only.

---

5. Strategic Synthesis

V1 (`crossAnalysisSynthesis`) — Rich messaging framework:

```
{
  keyInsights: Array<{
    insight: string
    source: 'industryMarketOverview' | 'icpAnalysisValidation' | ...  // ← V2 MISSING
    implication: string
    priority: 'high' | 'medium' | 'low'
  }>
  recommendedPositioning: string
  positioningStrategy: { primary, alternatives, differentiators, avoidPositions }
  messagingFramework: {
    coreMessage: string                // ← V2: optional
    supportingMessages: string[]       // ← V2: optional
    proofPoints: string[]              // ← V2: optional
    tonalGuidelines: string[]          // ← V2: optional
    adHooks: Array<{                   // ← V2 MISSING entirely
      hook: string
      technique: 'controversial' | 'revelation' | 'myth-bust' | ...  // 11 types
      targetAwareness: 'unaware' | 'problem-aware' | 'solution-aware' | ...
      source?: { type: 'extracted' | 'inspired' | 'generated', competitors?, platform? }
    }>
    angles: Array<{                    // ← V2 MISSING entirely
      name: string
      description: string
      targetEmotion: string
      exampleHeadline: string
    }>
    proofPointsDetailed: Array<{       // ← V2 MISSING entirely
      claim: string
      evidence: string
      source?: string
    }>
    objectionHandlers: Array<{         // ← V2 MISSING entirely
      objection: string
      response: string
      reframe: string
    }>
  }
  recommendedPlatforms: Array<{
    platform: 'Meta' | 'LinkedIn' | 'Google' | 'YouTube' | 'TikTok'  // ← V2: free string
    reasoning: string
    priority: 'primary' | 'secondary' | 'testing'
  }>
  criticalSuccessFactors: string[]
  potentialBlockers: string[]
```

```
  nextSteps: string[]
}
```

V2 (`strategicSynthesis`) — Core structure preserved but messaging framework gutted:

Gap Analysis:

Lost from V1	Impact	V3 Recommendation
`keyInsights[].source` field	Loses traceability to source section	**Restore** — enables "drill into source" UI
`messagingFramework.adHooks`	Loses 5-8 ready-to-use hooks with technique + awareness level	**MUST restore** — this IS the deliverable
`messagingFramework.angles`	Loses creative angles with emotions + example headlines	**MUST restore** — drives creative briefs
`messagingFramework.proofPointsDetailed`	Loses claim-evidence pairs	**Restore** — drives proof-based ads
`messagingFramework.objectionHandlers`	Loses objection-response-reframe triples	**Restore** — drives retargeting copy
`platform` enum → free string	Loses UI rendering precision	**Restore enum**
`messagingFramework` all-optional	Could produce empty framework	**Make required**

This is the second biggest regression. The messaging framework is the most actionable part of the entire pipeline.

---

6. Keyword Intelligence

V1 (`keywordIntelligence`) — Full SpyFu data model:

```
{
  clientDomain: {                      // ← V2 MISSING
    domain: string
    organicKeywords: number
    paidKeywords: number
    monthlyOrganicClicks: number
    monthlyPaidClicks: number
    organicClicksValue: number
    paidClicksValue: number
  }
  competitorDomains: DomainKeywordStats[]  // ← V2 MISSING
  organicGaps: KeywordOpportunity[]    // ← V2 MISSING
  paidGaps: KeywordOpportunity[]       // ← V2 MISSING
  sharedKeywords: KeywordOpportunity[] // ← V2 MISSING
  relatedExpansions: KeywordOpportunity[]  // ← V2 MISSING
  clientStrengths: KeywordOpportunity[]
  competitorTopKeywords: Array<{       // ← V2 MISSING
    competitorName: string
    domain: string
    keywords: KeywordOpportunity[]
  }>
  quickWins: KeywordOpportunity[]
  longTermPlays: KeywordOpportunity[]  // ← V2 MISSING
  highIntentKeywords: KeywordOpportunity[]
  contentTopicClusters: Array<{
    theme: string
    keywords: string[]
```

```
    searchVolumeTotal: number          // V1: number (V2: string | number)
    recommendedFormat: string          // ← V2 MISSING
  }>
  strategicRecommendations: {          // ← V2 MISSING entirely
    organicStrategy: string[]
    paidSearchStrategy: string[]
    competitivePositioning: string[]
    quickWinActions: string[]
  }
  metadata: {
    clientDomain: string               // V1: full metadata
    competitorDomainsAnalyzed: string[]
    totalKeywordsAnalyzed: number
    spyfuCost: number
    collectedAt: string
  }
  seoAudit?: {                         // ← V2 MISSING entirely
    technical: { pages[], sitemapFound, robotsTxtFound, overallScore }
    performance: { mobile: PageSpeedMetrics, desktop: PageSpeedMetrics }
    overallScore: number
  }
}
```

V2 (`keywordIntel`): Only keeps `keywords`, `quickWins`, `highIntentKeywords`, `clientStrengths`, `contentTopicClusters`, and `metadata.totalKeywordsAnalyzed`.

Gap Analysis:

Lost from V1	Impact	V3 Recommendation
`clientDomain` stats	Loses baseline organic/paid position	**Restore** — benchmarking input
`competitorDomains` stats	Loses competitor keyword strength comparison	**Restore** — competitive landscape
`organicGaps`, `paidGaps`	Loses gap-specific keyword lists	**Restore** — drives search strategy
`longTermPlays`	Loses high-difficulty opportunity pipeline	**Restore** — content strategy input
`strategicRecommendations`	Loses 4-pillar actionable strategy	**MUST restore** — makes keywords actionable
`seoAudit`	Loses technical SEO + PageSpeed data	**Restore** — landing page optimization input
`contentTopicClusters.recommendedFormat`	Loses format recommendations per cluster	**Restore** — content calendar input
Typed `KeywordOpportunity`	V1 has `searchVolume: number`, `cpc: number`, `difficulty: number`	**Restore strict types**

---

7. Media Plan

V1 (separate 10-section pipeline) — Massive structured output:

```
{
  executiveSummary: { overview, primaryObjective, budget, timeline, priorities }
  platformStrategy: Array<{
    platform, rationale, budgetPercentage, monthlySpend,
    campaignTypes, targetingApproach, expectedCplRange: { min, max },
    priority, adFormats, placements, synergiesWithOtherPlatforms,
    competitiveDensity, audienceSaturation, platformRiskFactors,
```

```
    qvcScore, qvcBreakdown: { targetingPrecision, leadQuality, costEfficiency, ... }
  }>
  icpTargeting: {
    segments: Array<{ name, description, targetingParameters, estimatedReach, funnelPosition }>
    platformTargeting: Array<{ platform, interests, jobTitles, customAudiences, exclusions }>
    demographics, psychographics, geographicTargeting
  }
  campaignStructure: {
    campaigns: Array<{ name, objective, platform, funnelStage, dailyBudget, adSets }>
    namingConvention: { campaignPattern, adSetPattern, adPattern, utmStructure }
    retargetingSegments, negativeKeywords
  }
  creativeStrategy: {
    angles, formatSpecs, testingPlan, refreshCadence, brandGuidelines
  }
  budgetAllocation: {
    totalMonthlyBudget, platformBreakdown, dailyCeiling,
    rampUpStrategy, funnelSplit, monthlyRoadmap
  }
  campaignPhases: Array<{ name, phase, durationWeeks, objective, activities, successCriteria }>
  kpiTargets: Array<{ metric, target, timeframe, measurementMethod, benchmark, scenarioThresholds }>
  performanceModel: { cacModel: { targetCAC, targetCPL, rates, LTV ratio }, monitoringSchedule }
  riskMonitoring: { risks: Array<{ risk, category, severity, mitigation, contingency }>, assumptions }
}
```

V2 (`mediaPlan`) — Collapsed to 5 fields:

```
{
  allocations: Array<{ channel, percentage, spend?, rationale? }>
  totalBudget: string
  timeline: string[]
  kpis: Array<{ channel?, metric?, target?, value? }>
  testingPlan: string[]
}
```

Gap Analysis:

Lost from V1	Impact	V3 Recommendation
`executiveSummary`	Loses high-level strategy overview	**Restore**
`platformStrategy` (full structure)	Loses CPL ranges, QVC scores, saturation, risk factors	**MUST restore** — core media plan
`icpTargeting`	Loses platform-specific targeting parameters	**MUST restore** — execution-ready targeting
`campaignStructure`	Loses naming conventions, UTM, retargeting, negatives	**MUST restore** — execution blueprint
`creativeStrategy`	Loses angles, format specs, refresh cadence	**Restore** — creative brief
`budgetAllocation` (structured)	Loses funnel split, ramp strategy, monthly roadmap	**Restore** — budget execution plan
`campaignPhases`	Loses phased rollout plan with go/no-go gates	**Restore** — implementation timeline
`kpiTargets` (structured)	Loses benchmarks, measurement methods, scenario thresholds	**MUST restore** — accountability framework
`performanceModel`	Loses CAC model + monitoring schedule	**Restore** — ROI projection

`riskMonitoring`	Loses risk matrix + assumptions	**Restore** — defensive planning

This is the most severe regression. V1 media plan was 10 structured sections; V2 is 5 flat fields.

Summary: V2 Schema Regression Severity

Section	V1 Fields	V2 Fields	Coverage	Severity
Industry & Market	6 objects, ~25 fields	6 objects, ~15 fields	~60%	MODERATE
Competitor Intel	7 objects, ~50+ fields	6 objects, ~15 fields	~30%	**CRITICAL**
ICP Validation	12 objects, ~80+ fields	12 fields (flat)	~15%	**CRITICAL**
Offer Analysis	5 objects, ~25 fields	7 objects, ~28 fields	~105%	GOOD (improved)
Strategic Synthesis	7 objects, ~40+ fields	7 objects, ~20 fields	~50%	**HIGH**
Keyword Intelligence	12 objects, ~50+ fields	6 objects, ~15 fields	~30%	**CRITICAL**
Media Plan	10 sections, ~100+ fields	5 flat fields	~5%	**CRITICAL**

V3 Elevation Priorities

Tier 1: MUST RESTORE (Critical regressions)

1. **ICP** `customerPsychographics` — Goals, fears, perceived enemy, day-in-the-life narrative
2. **ICP** `sensitivityAnalysis` — Best/base/worst case CAC projections + break-even
3. **ICP** `marketReachability` — Meta/LinkedIn/Google volume booleans
4. **Synthesis** `messagingFramework.adHooks` — Ready-to-use hooks with technique + awareness level
5. **Synthesis** `messagingFramework.angles` — Creative angles with target emotion + example headline
6. **Competitor** `threatAssessment` — 5-factor scoring per competitor
7. **Media Plan full structure** — Restore at minimum: `platformStrategy`, `campaignStructure`, `kpiTargets`, `budgetAllocation`

Tier 2: SHOULD RESTORE (High-value losses)

8. **ICP** `segmentSizing` — TAM/contacts per segment with priority factors
9. **ICP** `economicFeasibility` — Budget/TAM/CAC alignment
10. **ICP** `triggerEvents` **structured** — Frequency, urgency, detection method, recommended hook
11. **Competitor** `funnelBreakdown` **structured** — Landing page patterns, headline formulas, CTA hierarchy
12. **Competitor** `whiteSpaceGaps` **structured** — Exploitability × impact scoring
13. **Keyword** `strategicRecommendations` — 4-pillar actionable keyword strategy
14. **Keyword** `clientDomain` + `competitorDomains` — Baseline domain keyword stats

Tier 3: NICE TO HAVE (Quality improvements)

15. **Industry** `macroRisks` — Regulatory, downturn, consolidation context
16. **Competitor** `reviewData` — Trustpilot/G2 sentiment
17. **Keyword** `seoAudit` — Technical SEO + PageSpeed
18. **All sections**: Restore enum types where V1 used them (strings → enums)
19. **Synthesis** `keyInsights[].source` — Traceability to source section

V2-Only Additions to Keep

V2 Addition	Section	Value
`marketSize`	industryResearch	Good for sizing context
`adCount`	competitorIntel	Quick competitive pressure metric
`keywordOverlap`	competitorIntel	Novel competitive metric

`strengths`, `weaknesses`	offerAnalysis	Good additions
`pricingComparison`	offerAnalysis	Useful for pricing strategy
`opportunities` per competitor	competitorIntel	Actionable per-competitor

---

Implementation Note

V2's simplified schemas were designed for **streaming-friendly progressive rendering** in the chat UI. V1's rich schemas were designed for **batch generation** into a full document.

V3 should bridge both: Keep the streaming-friendly flat types for the real-time card rendering, but add a `detailed` nested object per section that populates after streaming completes. This gives users immediate visual feedback (cards) while building toward V1-depth output behind the scenes.

```
// V3 pattern: streaming-friendly base + rich details
type V3SectionOutput<TBase, TDetailed> = TBase & {
  detailed?: TDetailed  // Populated after streaming, before persist
}
```

Part 3: V3 Schema Elevation Strategy

The Core Problem

V2 has **more powerful tools than V1** (8 betaZodTool wrappers, web search, chart generation, live ad platform data) but **produces dramatically less structured output**. The runners underutilize available tools:

Runner	Tools Available	Tools Actually Used
`industry.ts`	9 tools	web_search only
`competitors.ts`	9 tools	web_search + adLibrary + spyfu + pagespeed
`icp.ts`	9 tools	web_search only
`offer.ts`	9 tools	web_search + firecrawl
`synthesize.ts`	9 tools	chartTool only
`keywords.ts`	9 tools	spyfu only
`media-planner.ts`	9 tools	googleAds + metaAds + ga4 + chart

V3 Elevation: Per-Section Plan

1. Industry Research → V3

Current: 8K tokens, web_search only, ~15 fields **V3 Target**: 12K tokens, web_search + firecrawl, ~25 fields

Restore from V1:

- `categorySnapshot.buyingBehavior` enum (`impulsive` | `committee_driven` | `roi_based` | `mixed`)
- `marketDynamics.macroRisks` object (regulatory, downturn, consolidation)
- `psychologicalDrivers` with `{ driver, description }` objects
- `audienceObjections` with `{ objection, howToAddress }` objects
- Make `category` , `marketMaturity` , `awarenessLevel` REQUIRED
- Restore enum types: `marketMaturity: 'early' | 'growing' | 'saturated'`

NEW for V3 (beyond V1):

- `marketBenchmarks` : { avgCPC, avgCPL, avgCAC } sourced from web search for the specific industry
- `trendSignals` : Array of { trend, direction: 'rising' | 'declining' | 'stable', evidence }
- `seasonalityCalendar` : 12-month array of { month, intensity: 1-10, notes }
- Add `firecrawl` tool to runner — scrape industry report landing pages for deeper data

Token budget: Increase to 12K (from 8K) — additional 4K covers restored + new fields

---

2. Competitor Intel → V3

Current: 8K tokens, web_search + adLibrary + spyfu + pagespeed, ~15 fields **V3 Target**: 16K tokens, same tools + firecrawl, ~50+ fields

Restore from V1:

- `threatAssessment` per competitor (5-factor scoring 1-10 + `topAdHooks` , `likelyResponse` , `counterPositioning`)
- `offer` , `funnels` , `adPlatforms` per competitor
- `pricingTiers` per competitor (from firecrawl scraping)
- `creativeLibrary.creativeFormats` as structured booleans
- `funnelBreakdown` structured (landing page patterns, headline structure, CTA hierarchy, form friction enum)
- `whiteSpaceGaps` as structured objects (gap, type enum, evidence, exploitability 1-10, impact 1-10, recommendedAction)

NEW for V3:

- `estimatedMonthlyAdSpend` per competitor (from SpyFu paid keywords data)
- `adCreativeThemes` : Array of { theme, count, platforms, exampleHook } from Ad Library
- `websitePerformance` per competitor: { performanceScore, lcp, cls } from PageSpeed
- Add `firecrawl` to competitor runner — scrape pricing pages for `pricingTiers`

Token budget: Increase to 16K (from 8K) — this is the richest section

---

3. ICP Validation → V3

Current: 8K tokens, web_search only, ~12 flat fields **V3 Target**: 16K tokens, web_search + firecrawl, ~80+ fields (V1 parity + enhancements)

Restore ALL from V1:

- `coherenceCheck` (5 booleans: clearlyDefined, reachable, adequateScale, hasPain, hasBudget)
- `painSolutionFit` (primaryPain, offerComponent, fitAssessment enum, notes)
- `marketReachability` (metaVolume, linkedInVolume, googleSearchDemand, contradictingSignals)
- `economicFeasibility` (hasBudget, purchasesSimilar, tamAlignedWithCac, notes)
- `riskScores` full matrix (risk, category enum, probability 1-5, impact 1-5, mitigation, contingency)
- `finalVerdict` (status: validated | workable | invalid, reasoning, recommendations)
- `customerPsychographics` (goalsAndDreams, fears, embarrassingSituations, perceivedEnemy, failedSolutions, dayInTheLife)
- `triggerEvents` structured (event, annualFrequency, urgencyLevel enum, detectionMethod, recommendedHook)
- `segmentSizing` (TAM accounts, contacts, segmentShare%, priorityTier, budgetWeight, priorityFactors 1-10)
- `samEstimate` (filteringFunnel, estimatedSAM, estimatedACV, confidence, dataSources)
- `sensitivityAnalysis` (bestCase, baseCase, worstCase CAC projections + breakEven)

NEW for V3:

- `platformAudienceSize` : { meta: number, linkedin: number, google: number } — estimated from web search benchmarks
- `buyingCommittee` : Array of { role, influence: 'decision-maker' | 'influencer' | 'gatekeeper', messagingAngle }
- Add `firecrawl` tool — scrape LinkedIn company pages and competitor case studies for ICP validation data

Token budget: 16K required — this is the deepest analytical section

---

4. Offer Analysis → V3

Current: 8K tokens, web_search + firecrawl, ~28 fields **V3 Target**: 10K tokens, same tools, ~35 fields

Keep V2 as-is (already at ~105% V1 coverage), add:

- `competitivePricing` : Array of { competitor, lowestTier, highestTier, comparison: 'cheaper' | 'similar' | 'premium' }
- `conversionPotential` : { landingPageScore: 1-10, urgencyFactors, frictionPoints }
- Tighten firecrawl usage — scrape client + top 2 competitor pricing pages

Token budget: 10K (from 8K) — modest increase

---

5. Strategic Synthesis → V3

Current: 10K tokens, chartTool only, ~20 fields **V3 Target**: 16K tokens, chartTool, ~40+ fields

Restore from V1:

- `keyInsights[].source` field (traceability to source section)
- `messagingFramework` as REQUIRED (not optional)
- `messagingFramework.adHooks` : Array of { hook, technique enum (11 types), targetAwareness enum (5 levels), source }
- `messagingFramework.angles` : Array of { name, description, targetEmotion, exampleHeadline }
- `messagingFramework.proofPointsDetailed` : Array of { claim, evidence, source? }
- `messagingFramework.objectionHandlers` : Array of { objection, response, reframe }
- `recommendedPlatforms[].platform` as enum (Meta | LinkedIn | Google | YouTube | TikTok)

NEW for V3:

- `budgetAllocationRecommendation` : { platform: string, percentage: number, reasoning: string }[]
- `competitivePositioningMap` : chart URL from chartTool (radar chart of client vs competitors)
- `channelPriorityMatrix` : chart URL (bar chart of platform recommendations)
- Force chartTool usage for at least 2 visualizations

Token budget: 16K (from 10K) — messaging framework needs space

---

6. Keyword Intelligence → V3

Current: 4K tokens, spyfu only, ~15 fields **V3 Target**: 10K tokens, spyfu + pagespeed, ~50+ fields

Restore from V1:

- `clientDomain` stats (organic/paid keywords, clicks, click value)
- `competitorDomains` stats (same per competitor)
- `organicGaps` , `paidGaps` as typed `KeywordOpportunity[]`
- `longTermPlays` keyword list
- `strategicRecommendations` (organicStrategy, paidSearchStrategy, competitivePositioning, quickWinActions)
- `contentTopicClusters[].recommendedFormat`
- Typed `KeywordOpportunity` : { keyword, searchVolume: number, cpc: number, difficulty: number, source enum }

NEW for V3:

- `seoAudit` from PageSpeed tool — add pagespeed to keyword runner
- `negativeKeywords` : Array of { keyword, reason } — keywords to exclude from paid campaigns
- `competitorKeywordGapChart` : chart URL from chartTool (Venn diagram of keyword overlap)
- Restore full metadata: `clientDomain` , `competitorDomainsAnalyzed` , `collectedAt`

Token budget: 10K (from 4K) — 2.5x increase for restored depth

---

7. Media Plan → V3

Current: 10K tokens, googleAds + metaAds + ga4 + chart, 5 flat fields **V3 Target**: 20K tokens, same tools, ~100+ fields (restore full V1 10-section structure)

Restore full V1 structure:

- `executiveSummary` (overview, primaryObjective, budget, timeline, priorities)
- `platformStrategy` per platform (rationale, budgetPercentage, monthlySpend, campaignTypes, targetingApproach, expectedCplRange, adFormats, placements, synergies, QVC score)
- `icpTargeting` (segments with targeting parameters, platform-specific audiences)

- `campaignStructure` (campaigns with ad sets, naming convention, UTM structure, retargeting, negatives)
- `creativeStrategy` (angles, format specs, testing plan, refresh cadence)
- `budgetAllocation` (platformBreakdown, dailyCeiling, rampUpStrategy, funnelSplit, monthlyRoadmap)
- `campaignPhases` (3-4 phases with objectives, activities, success criteria, go/no-go)
- `kpiTargets` (metric, target, timeframe, measurementMethod, benchmark, scenarioThresholds)
- `performanceModel` (CAC model, monitoring schedule)
- `riskMonitoring` (risks with severity, mitigation, contingency + assumptions)

NEW for V3:

- `liveDataInsights` : { platform, metric, currentValue, recommendation } — from Google Ads/Meta/GA4 when connected
- `budgetAllocationChart` : pie chart URL from chartTool
- `channelPerformanceForecast` : chart URL (projected metrics by month)
- `implementationChecklist` : Array of { task, platform, priority, estimatedHours }

Token budget: 20K (from 10K) — needs full 10-section structure

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New SDK Features to Leverage

Feature	Where to Use	Impact
Citations API	All runners → synthesis	Auto-cite web search sources with exact URLs
Output.object + Tools	Refactor runners to use `generateText` + `Output.object`	Unified tool calling + structured output
Web Search (`web_search_20250305`)	Already using; increase `max_uses` per runner	More comprehensive research
Batch API	"Refresh research" feature	50% cost savings for background re-runs
Token Counting	Pre-dispatch validation	Route to cheaper models when feasible
Extended Thinking	Synthesis + ICP runners	Force deep reasoning for CAC projections
input_examples	Complex tools (firecrawl, spyfu)	72% → 90% accuracy on tool parameters

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Token Budget Summary

Section	V2 Current	V3 Target	Delta
Industry Research	8K	12K	+4K
Competitor Intel	8K	16K	+8K
ICP Validation	8K	16K	+8K
Offer Analysis	8K	10K	+2K
Strategic Synthesis	10K	16K	+6K
Keyword Intelligence	4K	10K	+6K
Media Plan	10K	20K	+10K
Total	**56K**	**100K**	**+44K**

Cost impact: ~$0.15-0.30 more per full research run (at Sonnet 4.6 pricing).

Part 4: Current Onboarding UX Map

User Journey: Step by Step

Step 0: Landing Page (`/`)

- Hero: "Launch Your SaaS Marketing With AI Precision"
- CTA: "Generate Strategic Research" → `/journey`
- Auth: Clerk middleware gates `/journey` (public: `/`, sign-in, sign-up)

Step 1: Welcome State (First Visit to `/journey`)

Three-panel AppShell layout:

- Left: Collapsed sidebar
- Center: Welcome screen
- Right: Hidden (appears later with research progress)

User sees:

1. Three-step visual guide:
    - "1. Seed context" — give EGOS your homepage
    - "2. Verify findings" — review what AI discovered
    - "3. Watch research stream" — six sections generate live
2. Two input fields:
    - Website URL (required feel, optional technically)
    - LinkedIn company page (optional)
3. Two actions:
    - **"Analyze website first"** → triggers prefill (Firecrawl scrape + LLM extraction)
    - **"Start without website analysis"** → skips to chat

Step 2: Prefill Review (if website analyzed)

Card: "I pulled these details from your site"

- For each discovered field: label, confidence badge, source badge, value
- Per-field actions: Use this | Edit | Reject | View source
- Bottom: "Use these details and continue" | "Skip review"

Auto-discovered fields (from URL): companyName, industry, targetCustomers, targetJobTitles, companySize, headquartersLocation, productDescription, coreFeatures, valueProposition, pricing, competitors, uniqueDifferentiator, marketProblem, customerTransformation

Step 3: Session Creation

- POST `/api/journey/new-session` → Supabase `journey_sessions` row
- Session ID stored in localStorage

Step 4: Chat Begins

- Welcome message appears
- Profile Card shows: "What I know so far — X/8 essentials confirmed"
- Agent starts guided questioning flow

Step 5-10: Guided Onboarding Phases

Phase 1 — Company Basics (if not prefilled):

- Company name, website URL, business model (chips: B2B SaaS | B2C | Marketplace | Agency | Other)
- If URL provided: `scrapeClientSite(url)` fires → shows findings → asks for confirmation

Phase 2 — ICP Deep Dive:

- "Tell me about your best customers" → `primaryIcpDescription`
- Company size (multiSelect chips)
- Industry, job titles, geography, buying triggers
- Best client sources (multiSelect: Referrals | LinkedIn | Paid Ads | SEO | Events...)

TRIGGER: Once `businessModel` + `primaryIcpDescription` confirmed → `researchIndustry` fires immediately while agent keeps asking questions

Phase 3 — Product & Offer:

- Product description, core deliverables, pricing tiers, value prop
- Current funnel type (multiSelect: Lead Form | Free Trial | Webinar | Demo...)

Phase 4 — Competitive Landscape:

- "Who do you compete with?" → top 2-3 competitors
- **TRIGGER**: `competitorFastHits(competitorUrl)` fires for each competitor mentioned
- Unique edge, competitor frustrations, market bottlenecks

Phase 5 — Customer Journey:

- Situation before buying, desired transformation, common objections
- Sales cycle length (chips: <7d | 7-14d | 14-30d | 30d+)

Phase 6 — Brand & Budget:

- Brand positioning, monthly ad budget (**TRIGGER**: unlocks mediaPlan)
- Campaign duration, target CPL/CAC, goals (multiSelect)

Research Streaming (concurrent with Steps 5-10)

Research fires as triggers are met — NOT sequential with questions:

1. `industryResearch` → after businessModel + ICP description
2. `competitorIntel` → after industry complete + competitors named
3. `icpValidation` → after industry complete + ICP description
4. `offerAnalysis` → after competitors complete + product description
5. `strategicSynthesis` → after all 4 above complete
6. `keywordIntel` → after synthesis complete
7. `mediaPlan` → after synthesis + keywords + budget collected

Right panel shows live progress tracker: 7 sections with status dots, elapsed timer, checkmarks.

Research cards appear inline in chat as they complete — user can Approve or Request Revision on each card.

Step 11: Completion

- Agent presents strategic narrative weaving all findings
- Confirmation askUser: "Looks good, let's go" | "I want to change something"
- On confirm → enters **Strategist Mode** (no more field collection, pure strategy conversation)

Onboarding UX Audit: What's Good vs. What Needs Work

What's Working Well

- Progressive research triggering (fires between questions, not after all collected)
- Chip/pill UI for categorical questions reduces friction
- Prefill from website analysis eliminates ~5 questions
- Three-panel layout provides context without overwhelming
- Research cards appear inline — user sees value accumulating
- Resume prompt preserves progress across sessions
- Profile card shows "what I know" — transparent progress tracking

What Needs Improvement

Issue	Impact	Recommendation
8 question phases before any research	Too many questions before first research fires	Fire `industryResearch` after just URL + goal (2 fields)
Prefill is all-or-nothing	User must review ALL fields before continuing	Progressive prefill — inject confirmed fields as agent asks
No "quick start" path	Every user goes through full 15-20 question flow	Add "Express mode" — URL + budget + goal → fire all research with defaults
Research progress only on desktop	Mobile users can't see research status	Add inline progress indicator in chat on mobile
No mid-conversation research preview	User can't peek at running research	Add "peek" button during streaming — show partial results
Phase ordering is rigid	Agent asks in fixed order even if user volunteers info early	System prompt should detect volunteered info and skip ahead
No time estimate	User doesn't know how long the full process takes	Show "~10 min to complete strategy" estimate
Late budget collection	Budget asked in Phase 6 — blocks mediaPlan until very end	Move budget to Phase 1 (it's one of the 3 essential fields)

Ideal V3 Onboarding Flow (Recommended)

Minute 0-1: User provides URL → instant Firecrawl scrape → "Smart Prefill" card with 10+ fields **Minute 1-2**: Three essential questions: Goal, Budget bracket, ICP description (chips + text) **Minute 2-3**: `industryResearch` + `icpValidation` fire immediately. Agent asks about competitors while research runs. **Minute 3-5**: User names competitors → `competitorIntel` fires. Agent asks about offer/product. **Minute 5-7**: `offerAnalysis` fires. User sees 4 research cards building simultaneously. **Minute 7-10**: All 4 prerequisites complete → `strategicSynthesis` → `keywordIntel` → `mediaPlan` cascade. **Minute 10-12**: Full strategy complete. User has been engaged throughout with progressive value.

Key principle: Every user input should visibly trigger research. The flywheel accelerates with each answer.